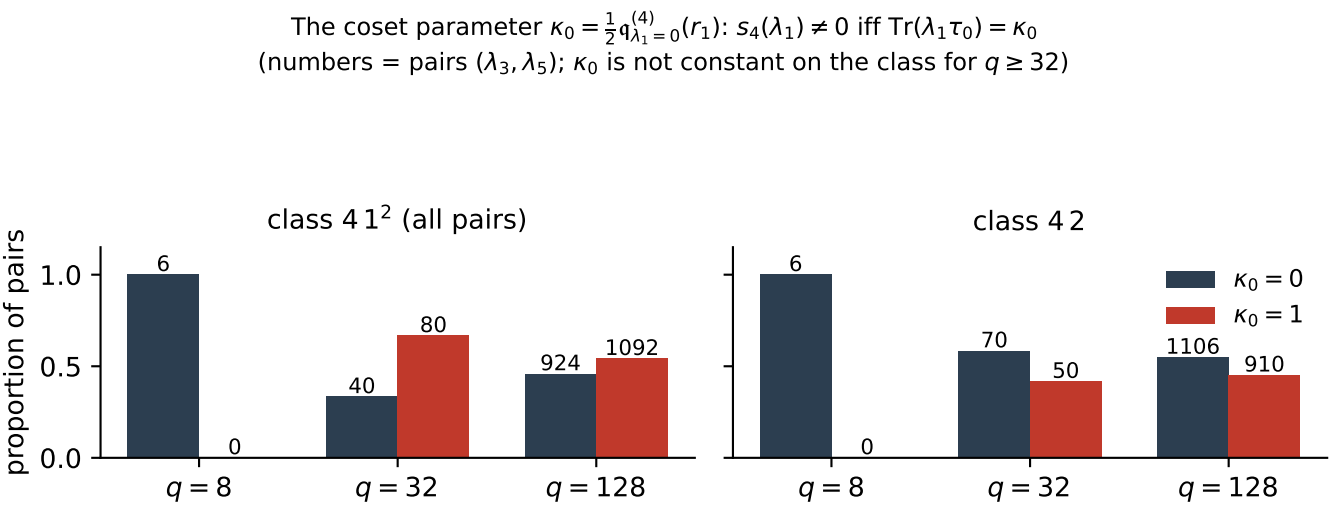
The coset parameter $\kappa_0 = \frac{1}{2}\mathfrak{q}^{(4)}_{\lambda_1=0}(r_1)$: $s_4(\lambda_1) \neq 0$ iff $\mathrm{Tr}(\lambda_1\tau_0) = \kappa_0$
(numbers = pairs (λ_3, λ_5); κ_0 is not constant on the class for $q \geq 32$)
class $4\,1^2$ (all pairs)
class $4\,2$
proportion of pairs
1.0
0.5
0.0
6
0
40
80
924
1092
6
0
70
50
1106
910
$\kappa_0 = 0$
$\kappa_0 = 1$
$q = 8$
$q = 32$
$q = 128$
$q = 8$
$q = 32$
$q = 128$